LAB v2 — JUNE 2026 Superseded by v3. Current figures are in the repository README.

Purple Team Home Lab

Detection Engineering Portfolio

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Overview

A self-built purple team lab covering the full attack-detect lifecycle end to end. For every attacker technique I execute the attack, build the detection, identify the telemetry gap, fix it, and re-test — the same loop a production detection-engineering team runs. Core thesis: **hardening ≠ detection**. Every prevention layer is bypassable, so detection is the safety net.

Infrastructure

Domain Controller	Windows Server 2022 — MARVEL.LOCAL Active Directory
Endpoints	2× Windows 10 (domain-joined)
Attacker	Kali Linux — Sliver C2, Impacket, BloodHound, kerbrute
SIEM	Splunk + Universal Forwarder + Sysmon on all Windows hosts

8-Phase Attack Chain — MITRE ATT&CK

#	PHASE	TECHNIQUE
1	Reconnaissance	`T1046 / T1087`
2	Password Spray	`T1110.003`
3	LOLBin Execution (InstallUtil)	`T1218.004`
4	Kerberoasting (SQLService)	`T1558.003`
5	Lateral Movement (Sliver C2, mTLS)	`T1071.001 / T1572`
6	Credential Dumping (Impacket secretsdump)	`T1003.002 / .006`
7	Persistence (Scheduled Task)	`T1053.005`
8	C2 Evasion + Sysmon gap fix	`Defense Evasion`

Phase 8 is the narrative climax — a detection gap found, diagnosed, and closed.

Detection Engineering Output

- **9 production-grade Sigma rules** — vendor-agnostic, convertible to SPL / KQL / EQL, version-controlled
- **30+ SPL queries** — detection cards, threat-hunt hypotheses, hardening verification
- **MITRE ATT&CK Navigator layer** — 18 techniques mapped for coverage visibility
- **Splunk dashboard** — 8-panel Simple XML, savedsearches.conf, props.conf, datamodels.conf

Highlighted Finding — C2 Beacon Detection Gap

Default Sysmon configuration silently missed the Sliver C2 beacon (mTLS, port 443, 60s + 30% jitter) originating from a user-writable path. I traced the gap to missing NetworkConnect monitoring, deployed a behavioral watchlist via GPO, and confirmed Sysmon EC3 then captured the beacon. Key insight: detect **behavior** (a process from a user-writable path beaconing on a periodic interval), not IOCs like filename or hash — which an attacker trivially changes.

Tooling

Detection: Splunk SPL, Sigma, MITRE ATT&CK Navigator, Sysmon | **Offensive:** Sliver C2, Impacket, BloodHound, kerbrute, SharpKatz, John the Ripper, Certipy | **Dev:** Python, Bash, PowerShell, Git, Java / Spring Boot